2.32 Mixed Integration Practice

Worked Solutions

1a)

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c.$$

1b)

$$\int \frac{1}{1+x^2}\,dx = \arctan x + c.$$

1c)

$$\int \frac{1}{\sqrt{x^2-1}}\,dx = \operatorname{arcosh} x + c.$$

1d)

$$\int \frac{1}{\sqrt{1+x^2}}\,dx = \operatorname{arsinh} x + c.$$

1e)

$$\int \frac{1}{1-x^2}\,dx = \operatorname{artanh} x + c.$$

2a)

$$\begin{aligned}\int_{\pi/4}^{\pi/2} 6\cot x\,dx = 6\int_{\pi/4}^{\pi/2} \frac{\cos x}{\sin x}\,dx &= 6\left[\ln|\sin x|\right]_{\pi/4}^{\pi/2} \\ &= 6\left(\ln 1 - \ln\frac{\sqrt{2}}{2}\right) = -6\ln\frac{\sqrt{2}}{2} = \ln 8.\end{aligned}$$

Thus $k = 8$.

2b)

The integral is improper because $\cot 0$ is undefined.

$$\int_0^{\pi/2} 6\cot x\,dx = \lim_{a\to 0^+}\int_a^{\pi/2} 6\cot x\,dx = \lim_{a\to 0^+}\left(6\ln\left|\sin\frac{\pi}{2}\right| - 6\ln|\sin a|\right).$$

As $a \to 0^+$, $\sin a \to 0$ and $\ln|\sin a| \to -\infty$. The integral diverges and is undefined.

3a)

Let $u = \tan x$, so $du = \sec^2 x\, dx$ and $dx = \cos^2 x\, du$. Then

$$\begin{aligned}\int \frac{1}{25\cos^2 x + \sin^2 x}\, dx &= \int \frac{\cos^2 x}{25\cos^2 x + \sin^2 x}\, du \\ &= \int \frac{1}{25 + \tan^2 x}\, du = \int \frac{1}{25 + u^2}\, du.\end{aligned}$$

3b)

With $u = \tan x$, the limits $x = 0$ and $x = \arctan 5$ become $u = 0$ and $u = 5$. Hence

$$\begin{aligned}\int_0^{\arctan 5} \frac{1}{25\cos^2 x + \sin^2 x}\, dx &= \int_0^5 \frac{1}{25 + u^2}\, du \\ &= \frac{1}{25}\int_0^5 \frac{1}{1 + (u/5)^2}\, du = \frac{1}{5}\left[\arctan\left(\frac{u}{5}\right)\right]_0^5 \\ &= \frac{1}{5}\left(\frac{\pi}{4} - 0\right) = \frac{\pi}{20}.\end{aligned}$$

Thus $k = \frac{1}{20}$.

4)

Let $u = x^2$, so $du = 2x\, dx$. The limits become $u = \ln 2$ and $u = \ln 5$.

$$\begin{aligned}I &= \int_{\sqrt{\ln 2}}^{\sqrt{\ln 5}} \frac{x}{\sinh(x^2)}\, dx = \frac{1}{2}\int_{\ln 2}^{\ln 5} \frac{1}{\sinh u}\, du \\ &= \frac{1}{2}\int_{\ln 2}^{\ln 5} \frac{2e^u}{e^{2u} - 1}\, du.\end{aligned}$$

Set $v = e^u$, so $du = dv/v$; the limits become 2 and 5.

$$\begin{aligned}I &= \int_2^5 \frac{1}{v^2 - 1}\, dv = \frac{1}{2}\int_2^5 \left(\frac{1}{v-1} - \frac{1}{v+1}\right) dv \\ &= \frac{1}{2}\left[\ln|v-1| - \ln|v+1|\right]_2^5 \\ &= \frac{1}{2}\left((\ln 4 - \ln 6) - (\ln 1 - \ln 3)\right) = \frac{1}{2}\ln 2.\end{aligned}$$

5a)

Complete the square:

$$x^2 - 12x + 61 = (x-6)^2 + 25.$$

Therefore

$$\int \frac{1}{x^2 - 12x + 61}\, dx = \frac{1}{25}\int \frac{1}{1 + ((x-6)/5)^2}\, dx = \frac{1}{5}\arctan\left(\frac{x-6}{5}\right) + c.$$

5b)

$$\begin{aligned}\int_5^7 \left(3 - \frac{78}{x^2 - 12x + 61}\right) dx &= \left[3x - \frac{78}{5}\arctan\left(\frac{x-6}{5}\right)\right]_5^7 \\ &= 6 - \frac{156}{5}\arctan\left(\frac{1}{5}\right).\end{aligned}$$

5c i)

At an x-intercept, $y = 0$:

$$0 = 3 - \frac{78}{x^2 - 12x + 61},$$
$$0 = x^2 - 12x + 35 = (x - 5)(x - 7).$$

Thus $x = 5$ or $x = 7$, so $P = (5, 0)$ and $Q = (7, 0)$.

5c ii)

$$\frac{dy}{dx} = \frac{156(x - 6)}{(x^2 - 12x + 61)^2}.$$

At a stationary point $dy/dx = 0$, so $x = 6$. Then

$$y\big|_{x=6} = 3 - \frac{78}{36 - 72 + 61} = 3 - \frac{78}{25} = -\frac{3}{25}.$$

Thus $R = (6, -3/25)$.

5c iii)

$$\int_5^6 \left(3 - \frac{78}{x^2 - 12x + 61}\right) dx = \left[3x - \frac{78}{5}\arctan\left(\frac{x - 6}{5}\right)\right]_5^6$$
$$= 3 - \frac{78}{5}\arctan\left(\frac{1}{5}\right).$$

The curve lies below the x-axis on this interval, so the exact area is

$$\frac{78}{5}\arctan\left(\frac{1}{5}\right) - 3 \quad \text{units}^2.$$

6a)

Write

$$\frac{2x^3 + x^2 + x}{(x + 1)(x^2 + 1)} = A + \frac{B}{x + 1} + \frac{Cx + D}{x^2 + 1}.$$

Equating coefficients gives $A = 2$, $B = -1$, $C = 0$, $D = -1$. Hence

$$\int \frac{2x^3 + x^2 + x}{(x + 1)(x^2 + 1)}\, dx = \int \left(2 - \frac{1}{x + 1} - \frac{1}{x^2 + 1}\right) dx$$
$$= 2x - \ln|x + 1| - \arctan x + c.$$

6b)

$$\int_0^1 \frac{2x^3 + x^2 + x}{(x + 1)(x^2 + 1)}\, dx = [2x - \ln|x + 1| - \arctan x]_0^1$$
$$= 2 - \ln 2 - \frac{\pi}{4}.$$

7a)

Complete the square: $4x^2+20x+34 = 9+4(x+5/2)^2$. Let $\sinh u = 2(x+5/2)/3$, so $dx = \frac{3}{2}\cosh u\,du$. At $x=-5/2$, $u=0$; at $x=-3/2$, $u=\operatorname{arsinh}(2/3)$. Therefore

$$I = \frac{1}{2}\int_0^{\operatorname{arsinh}(2/3)} du = \frac{1}{2}\operatorname{arsinh}\frac{2}{3} = \frac{1}{2}\ln\left(\frac{2+\sqrt{13}}{3}\right).$$

7b)

Complete the square: $-4x^2+4x+8 = 9-4(x-1/2)^2$. Let $\sin u = 2(x-1/2)/3$, so $dx = \frac{3}{2}\cos u\,du$. The limits become 0 and $\pi/6$.

$$I = \frac{1}{2}\int_0^{\pi/6} du = \frac{\pi}{12}.$$

7c)

Complete the square: $4x^2+12x+18 = 9+4(x+3/2)^2$. Let $\tan u = 2(x+3/2)/3$, so $dx = \frac{3}{2}\sec^2 u\,du$. The limits become $\pi/6$ and $\pi/4$.

$$I = \frac{1}{6}\int_{\pi/6}^{\pi/4} du = \frac{1}{6}\left(\frac{\pi}{4}-\frac{\pi}{6}\right) = \frac{\pi}{72}.$$

7d)

Complete the square: $-4x^2+28x-40 = 9-4(x-7/2)^2$. Let $\tanh u = 2(x-7/2)/3$, so $dx = \frac{3}{2}\operatorname{sech}^2 u\,du$. The limits $x=3,4$ give $u = -\operatorname{artanh}(1/3), \operatorname{artanh}(1/3)$, respectively. Thus

$$\begin{aligned} I &= \frac{1}{6}\int_{-\operatorname{artanh}(1/3)}^{\operatorname{artanh}(1/3)} du = \frac{1}{3}\operatorname{artanh}\frac{1}{3} \\ &= \frac{1}{6}\ln\left(\frac{1+1/3}{1-1/3}\right) = \frac{1}{6}\ln 2. \end{aligned}$$

8)

Factor the denominator and use partial fractions:

$$1-x^4 = (1-x)(1+x)(1+x^2), \qquad \frac{12x^2}{1-x^4} = \frac{3}{1-x}+\frac{3}{1+x}-\frac{6}{1+x^2}.$$

Therefore

$$\begin{aligned} I &= \left[-3\ln|1-x| + 3\ln|1+x| - 6\arctan x\right]_{-1/\sqrt{3}}^{1/\sqrt{3}} \\ &= 3\ln(2+\sqrt{3}) - \pi - \left(3\ln(2-\sqrt{3})+\pi\right) \\ &= 3\ln(7+4\sqrt{3}) - 2\pi. \end{aligned}$$

Thus $p=7$, $q=48$ and $r=-2$.